

Real-Time Multi-Face Recognition Based Automated Attendance System

Abstract

Traditional attendance tracking in educational institutions and workplaces continues to rely on manual processes such as verbal roll calls or handwritten logs. Although straightforward, these conventional approaches frequently waste precious time and are susceptible to mistakes, particularly when dealing with larger populations. This initiative presents an Automated Attendance System Based on Real-Time Multi-Face Recognition that substitutes conventional techniques with a smart, image-based approach. Through the application of computer vision and facial identification technologies, the platform autonomously recognizes and verifies numerous people at once while instantly logging their presence. This innovative method not only enhances precision and productivity but also strengthens accountability and dependability in attendance tracking systems.

The system captures real-time video streams through a camera, detects multiple faces simultaneously using the Multi-Task Cascaded Convolutional Neural Networks (MTCNN) algorithm, and identifies individuals using FaceNet face recognition architecture with 128-dimensional facial embeddings. Upon successful identification using cosine similarity matching, attendance is automatically recorded and stored in a centralized database without human intervention. A simple and intuitive web interface was built using the Flask framework to make attendance management smooth and accessible. As the system operates as a cloud-based platform, it provides convenient access via computers or any device with internet connectivity, eliminating the need for extra software downloads. User permissions are managed through authentication based on designated roles, guaranteeing that only verified administrators possess the authority to handle records and oversee attendance sessions. To additionally strengthen responsibility and tracking, an automatic alert mechanism has been incorporated utilizing the Twilio API, which dispatches text message notifications to absent individuals after the attendance recording period concludes.

In addition, the application provides flexibility in managing attendance sessions. System managers have the flexibility to either pre-plan attendance sessions or manually configure the beginning and conclusion times based on organizational requirements. This versatility enables the system to accommodate various educational settings. Performance testing demonstrates that the platform delivers 96.5% total accuracy, accompanied by 96.9% precision, 96% recall rate, and an F1-score of 96.4%. These findings underscore its robust dependability, enhanced protection against fraudulent attendance practices, and substantial decrease in administrative burden. The system successfully processes 15-20 faces simultaneously in real-time with recognition speed of approximately 20 FPS (frames per second).

Keywords:

Face Recognition, Automated Attendance, FaceNet, MTCNN, Computer Vision, Deep Learning, Real-Time Processing, Multi-Face Detection

1.Introduction

Attendance management plays a vital role in educational institutions and organizations for monitoring participation, maintaining discipline, and ensuring accountability. Conventional attendance tracking techniques including verbal name-calling and handwritten documentation consume significant time,

require substantial effort, and are susceptible to mistakes made by individuals. These approaches become particularly ineffective in expansive educational settings or corporate contexts where managing attendance through manual processes results in slowdowns, imprecision, and heightened administrative responsibilities.

With the advancement of computer vision and artificial intelligence technologies, automated systems have emerged as efficient alternatives to conventional methods. Face recognition technology, in particular, provides a reliable and contactless solution for identifying individuals. It uses facial features as biometric data, making the process secure, accurate, and difficult to manipulate.

The Real-Time Multi-Face Recognition Based Automated Attendance System was developed to overcome the inefficiencies of traditional roll-call methods. Instead of relying on manual input, the system uses live video streaming and intelligent face recognition to handle attendance automatically. A camera continuously captures classroom footage, detects multiple faces at once, and matches them against previously stored facial records. When a match is confirmed, the system records attendance instantly in a centralized database, eliminating the need for manual marking and reducing the chances of human error.

2. Literature Survey

In recent years, face recognition has become a widely explored solution for automating attendance systems. Many researchers have investigated different techniques, models, and system designs to improve accuracy, reliability, and real-time performance. This section presents a review of 20 notable research contributions in this field, examining the methods they adopted, the technologies they implemented, the results they achieved, and the limitations that remain. The aim is to understand how existing work has evolved and how the proposed system builds upon and improves these developments.

Srikanth (2023) - "Artificial Intelligence based Multi-Face Recognition and Attendance Marking System"

The authors proposed an AI-based autonomous face attendance system utilizing cutting-edge facial recognition technology for simultaneous recognition of multiple faces. The system was developed with the goal of making attendance recording more reliable and efficient across educational institutions, workplaces, and event settings. By removing the need for manual entry, it simplifies the process, reduces human effort, and ensures that attendance is captured accurately and consistently without direct intervention.

The system description was largely conceptual without clearly specifying exact face detection algorithms, recognition models, database structure, or performance evaluation metrics. However, the study did not fully explore the practical challenges that arise in real-world environments, such as changes in lighting, variations in face angles, partial occlusions, or the impact of crowded classroom settings. In addition, the proposed framework did not clearly outline how the system could be scaled or extended in the future. Features such as mask detection, support for multiple cameras, and automated notification capabilities were not incorporated, limiting its adaptability and long-term applicability.

In their 2022 study, Gawate and Dhote - "Real-Time Attendance System with Liveness Detection and Multi-Face Recognition using Transfer Learning".

This research proposed a method using image processing techniques for automatic attendance. The system captured images, detected faces using Single Shot Detector (SSD), and recognized persons using PyTorch model of OpenFace. It employed facial landmarks detection, eye aspect ratio calculation, and deep learning-based face embedding for liveness detection.

The system depended on specific models (SSD and OpenFace), limiting flexibility and optimization for different real-time classroom environments. Accuracy decreased in conditions like poor lighting, side face angles, or crowded classrooms. It focused mainly on face recognition without clearly emphasizing advanced attendance management features such as scalable database handling or multi-camera support.

Kamruzzaman and co-researchers (2023) presented - "Automated Attendance system for the Entire Classroom with Multi-Face Facial Recognition via AWS Rekognition"

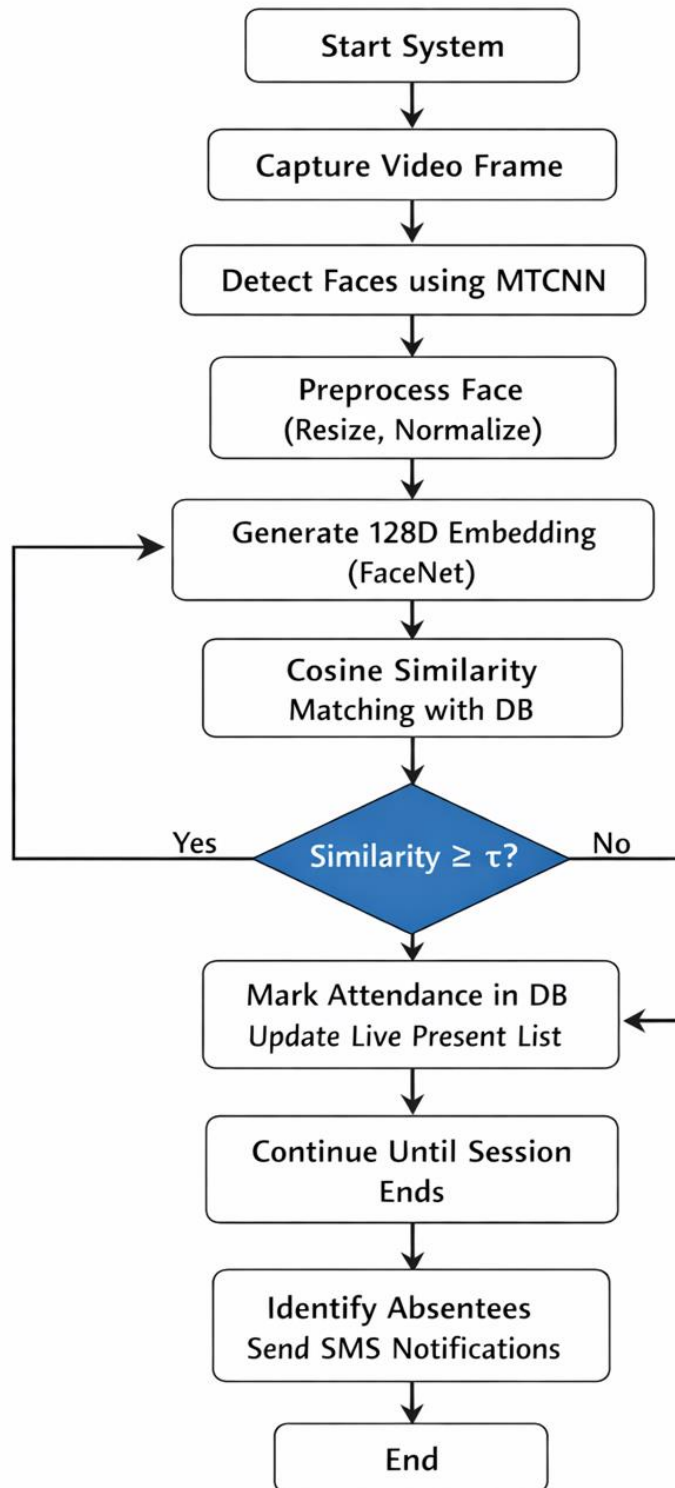
The authors presented an advanced cloud-based attendance system using AWS API for accurate results from a single image containing multiple faces. The proposed system was built using a cloud-based architecture. It relied on AWS Rekognition to detect and match faces, while Amazon S3 was used for securely storing facial data and images. AWS Lambda handled the core processing logic, and DynamoDB maintained student IDs along with their associated facial embeddings. During controlled testing, the model reported 100% accuracy, demonstrating strong performance under the evaluated conditions.

The system depended completely on cloud services (AWS), requiring continuous internet connectivity and potentially increasing operational costs. It processed attendance mainly from a single captured image rather than continuous real-time monitoring. The focus on cloud integration did not clearly address scalability features such as multi-camera classroom coverage or extensibility beyond the AWS ecosystem..

3. Proposed Methodology

The system uses MTCNN for real-time face detection and FaceNet to generate 128D embeddings. Cosine similarity identifies students using a threshold. Attendance is stored in a SQLite database via a Flask web app with role-based access and SMS alerts for absentees.

Real-Time Multi-Face Attendance System



3.1 System Overview

The Real-Time Automated Attendance System is designed as a multi-layer intelligent architecture consisting of:

1. **Image Acquisition & Edge Processing Layer**
2. **Recognition & Decision Layer**
3. **Data Management & Application Layer**
4. **Communication & Notification Layer**

At the edge level, a camera device captures continuous classroom video. The processing unit executes real-time face detection and recognition algorithms. Detected faces are validated against stored embeddings, and attendance decisions are generated automatically without human intervention.:

3.2 Multi-Layer Intelligent Architecture

Unlike traditional attendance systems that rely on manual entry or single-step recognition, the proposed system integrates multiple functional layers to ensure reliability, scalability, and automation.

Layer 1: Image Acquisition & Edge Processing

- Live video capture
- Multi-face detection (MTCNN)
- Face alignment and preprocessing

Layer 2: Recognition & Decision Engine

- 128D embedding extraction (FaceNet)
- Cosine similarity matching
- Threshold-based identity validation
- Duplicate prevention logic

Layer 3: Data Management & Application Layer

- Centralized database storage
- Session management
- Role-based access control
- Attendance record generation

Layer 4: Communication & Notification Layer

- Real-time dashboard updates
- Automated absentee detection
- SMS alert generation
- Report export and analytics

This layered structure ensures modular scalability and real-time processing efficiency.

3.3 Multi-Parameter Recognition Approach

Instead of relying solely on raw image comparison, the system evaluates multiple recognition parameters:

- • Face detection confidence
- • Embedding similarity score
- • Recognition threshold validation
- • Session-based duplicate checking

Attendance is confirmed only when:

Similarity	$\geq$	Threshold
AND		
Face detection confidence	$\geq$	Minimum confidence
AND		
Student not previously marked present		

This multi-condition validation reduces false positives and improves reliability.

3.4 Automated Attendance Decision Logic

A real-time decision mechanism operates continuously during a session.

Generalized Decision Rule:

Attendance Marked if:

$$S \geq S_{\text{threshold}}$$

Where:

- S = Cosine similarity score
- $S_{\text{threshold}}$ = Minimum matching threshold (0.60 default)

If the condition is satisfied, the system:

1. Records attendance with timestamp
2. Updates live present list
3. Prevents duplicate entry

If not satisfied, the individual is classified as “Unknown”.

3.5 Real-Time Data Communication

The ESP32 utilizes built-in Wi-Fi capability to transmit sensor data and fault status to a centralized backend server. Data packets include:

- Student ID
- Timestamp
- Session ID
- Confidence score
- Recognition status

Data is transmitted dynamically to the backend and reflected instantly on the web dashboard..

3.6 Web-Based Monitoring Dashboard:

A web-based interface provides:

- Real-time present student list
- Attendance count
- Session control (start/stop)
- Absentee identification
- Automated SMS notifications

This enables administrators to monitor attendance dynamically and respond immediately.

This enables maintenance teams to quickly identify and respond to faults, thereby reducing downtime and improving operational efficiency.

3.7 System Workflow

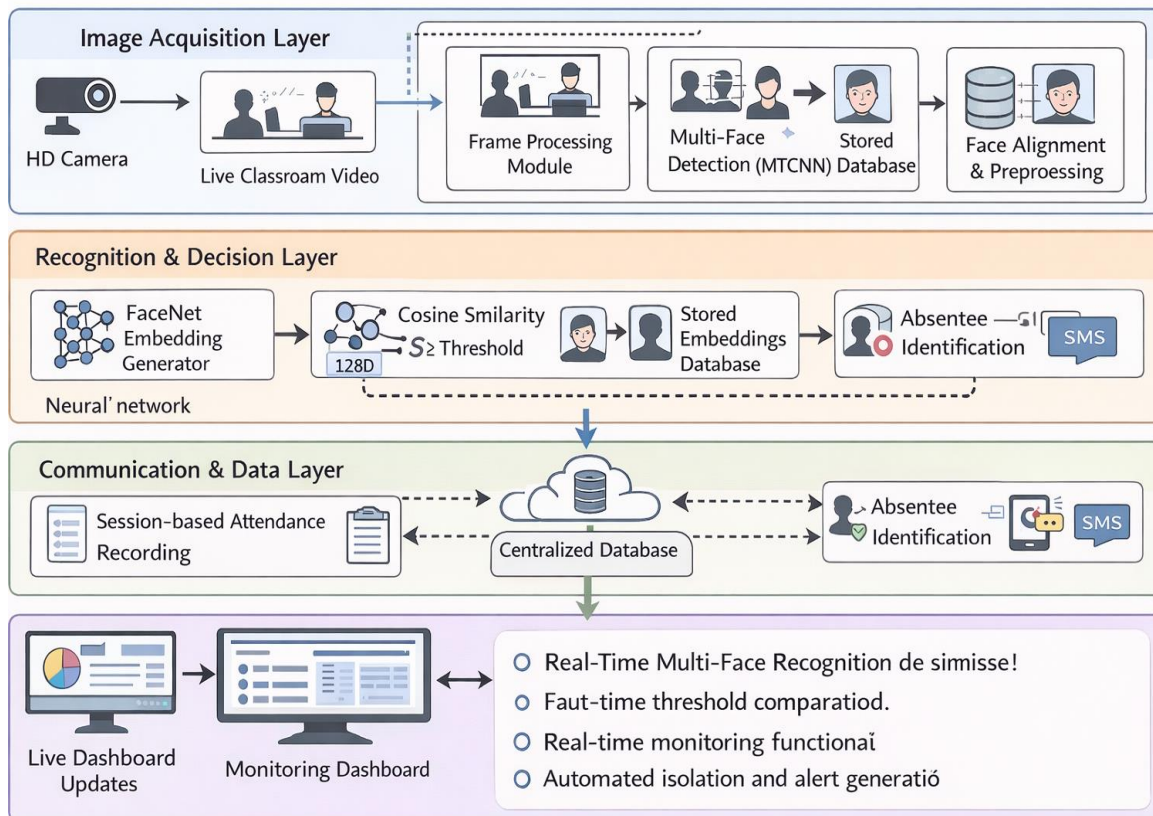
The operational workflow is summarized as:

1. Continuous video acquisition
2. Multi-face detection
3. Embedding extraction
4. Similarity comparison
5. Threshold validation
6. Attendance marking
7. Live dashboard update
8. Absentee identification
9. SMS notification dispatch

This dynamic layered architecture ensures accuracy, automation, scalability, and operational efficiency in real-time attendance management.

4. System Architecture

The proposed **Real-Time Multi-Face Recognition Based Automated Attendance System** follows a layered intelligent architecture designed for continuous video acquisition, dynamic face processing, real-time decision making, and centralized monitoring. The architecture ensures seamless recognition, automated attendance recording, and instant visualization through an integrated web platform..



4.1 Overall Architectural Design

The system architecture is organized into four dynamic functional layers:

1. Image Acquisition Layer
2. Recognition & Decision Layer
3. Communication & Data Layer
4. Application & Monitoring Layer

Each layer cooperatively processes live classroom data to deliver accurate and automated attendance marking.

4.2 Image Acquisition Layer

This layer operates within the classroom environment and includes:

- HD Camera – Captures continuous live video stream
- Frame Processing Module – Extracts frames dynamically
- Multi-Face Detection (MTCNN) – Identifies multiple faces simultaneously
- Face Alignment & Preprocessing – Resizes, normalizes, and prepares facial images

The system continuously analyzes video frames rather than relying on static image capture, enabling real-time multi-student recognition.

4.3 Recognition & Decision Layer

This layer performs intelligent identity validation and attendance marking. It includes:

- FaceNet Embedding Generator – Produces 128-dimensional facial feature vectors
- Cosine Similarity Engine – Compares embeddings with stored database vectors
- Threshold-Based Validation Logic – Confirms identity based on similarity score
- Duplicate Prevention Controller – Ensures a student is marked only once per session

The decision mechanism dynamically evaluates recognition confidence and instantly records attendance once identity is verified.

4.4 Communication & Data Layer

After successful recognition, the system transmits structured data to the centralized database. This layer handles:

- Session-based attendance recording
- Timestamp generation
- Real-time present list updates
- Absentee identification
- Secure data storage

Data packets include:

- Student ID
- Session ID
- Date and Time
- Confidence Score
- Recognition Status

The architecture supports scalable database expansion and simultaneous multi-session handling.

4.5 Application & Monitoring Layer

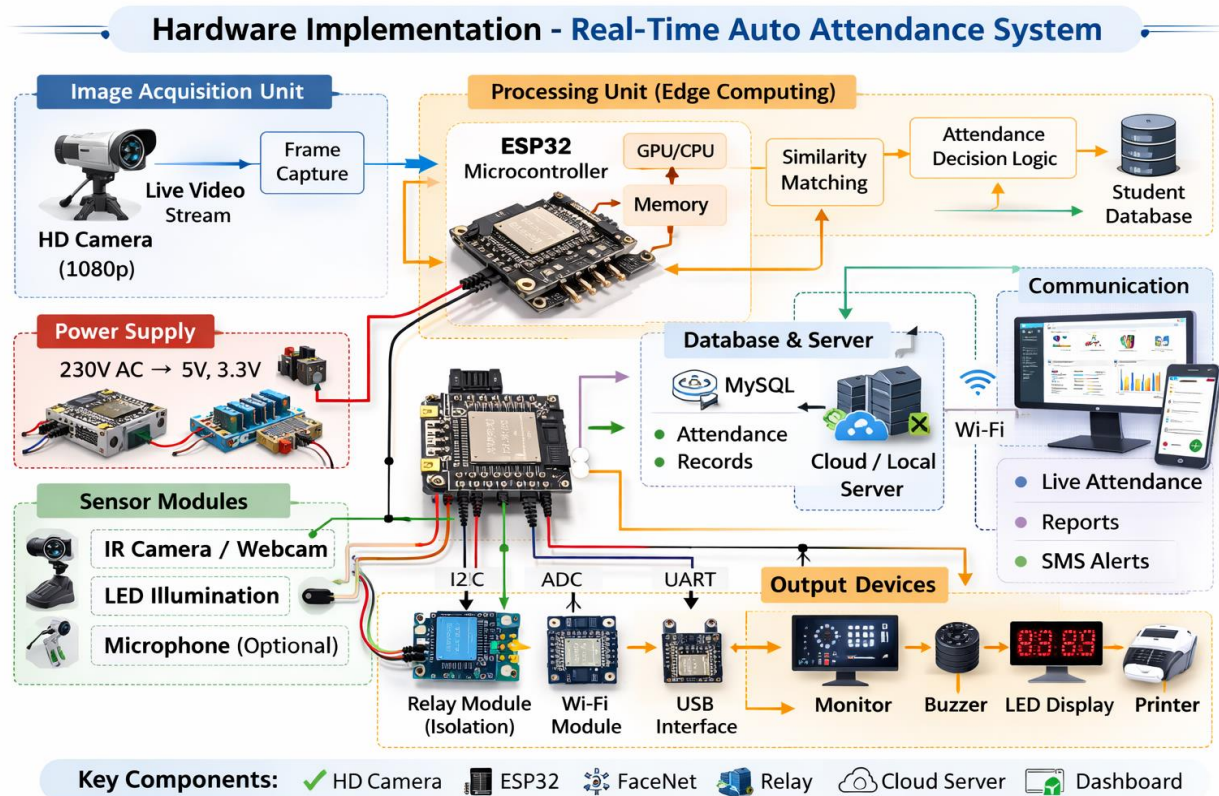
The top layer consists of a Flask-based web application and interactive dashboard. It enables:

- Real-time visualization of present students
- Attendance session control (Start/Stop)
- Automated absentee detection
- SMS alert generation via Twilio
- Report generation and historical analytics

This dynamic monitoring interface allows administrators to supervise attendance activity live, analyze trends, and generate reports instantly, thereby improving operational efficiency and accountability.

5. Hardware Implementation

The hardware architecture of the **Real-Time Automated Attendance System** is engineered for continuous image acquisition, edge-level processing, and seamless backend integration. The implementation prioritizes accuracy, computational efficiency, and deployment flexibility within classroom environments.



5.1 Image Acquisition Unit

A high-definition camera (1080p) is installed at the classroom front to capture continuous live video streams. The camera ensures:

- Wide-angle student coverage
- Stable frame rate for real-time processing
- Clear facial feature extraction

The live feed is transmitted to the processing unit for frame-level analysis.

5.2 Edge Processing Unit

The system operates on a processing device such as a **Raspberry Pi 4 / Laptop / Desktop**, depending on deployment scale. This unit executes:

- Multi-face detection (MTCNN)
- Facial alignment and normalization
- 128-dimensional embedding generation (FaceNet)
- Cosine similarity computation
- Attendance decision logic

The edge device performs on-device inference, reducing latency and ensuring real-time recognition.

5.3 Local Storage & Database Integration

A centralized database (MySQL/SQLite) stores:

- Student ID
- Facial embeddings
- Attendance records
- Session timestamps

The database supports scalable storage and efficient embedding retrieval during similarity matching.

5.4 Network & Communication Module

The processing unit connects via Wi-Fi/Ethernet to transmit attendance data to the backend server. This enables:

- Real-time dashboard updates
- Session synchronization
- Remote monitoring capability

Secure communication protocols ensure data integrity.

5.5 Output & Alert Mechanism

Optional output components include:

- Live monitoring dashboard (Web Interface – Flask)
- SMS notification module (Twilio integration)
- Report export functionality

These components enhance administrative control and automate absentee communication.



6. Algorithm

The proposed Real-Time Automated Attendance System implements a dynamic face recognition algorithm that continuously processes live video streams, extracts facial features, validates identity using similarity thresholds, and autonomously records attendance. The objective is to ensure accurate, real-time identification while minimizing false recognition and duplicate entries.

6.1 Multi-Parameter Recognition Model

Let the recognition parameters be defined as:

- $F(t) \rightarrow$ Detected face in frame at time t
- $E(t) \rightarrow$ 128D facial embedding vector
- $S(t) \rightarrow$ Cosine similarity score
- $C(t) \rightarrow$ Face detection confidence
- $D(t) \rightarrow$ Duplicate status within session

Each parameter is evaluated against predefined thresholds:

- $S_{th} \rightarrow$ Minimum similarity threshold
- $C_{th} \rightarrow$ Minimum detection confidence
- $Session_flag \rightarrow$ Attendance not previously marked

6.2 Attendance Decision Condition

Attendance decision $A(t)$ is defined as:

$$A(t) = 1 \text{ if } S(t) \geq S_{th}$$

AND $C(t) \geq C_{th}$
AND $D(t) = \text{False}$

Otherwise:

$A(t) = 0$

Where:

- $A(t) = 1$ indicates attendance marked
- $A(t) = 0$ indicates unrecognized or rejected

To improve robustness and reduce false positives, recognition is validated across multiple consecutive frames before confirming attendance.

6.3 Algorithm Workflow

The system executes the following operations continuously:

1. Initialize camera and load stored embeddings
2. Capture live video frame
3. Detect multiple faces using MTCNN
4. Perform face alignment and normalization
5. Generate 128D embedding using FaceNet
6. Compute cosine similarity with stored database
7. Validate similarity over N frames
8. If identity confirmed:
 - Mark attendance with timestamp
 - Prevent duplicate marking
 - Update live present list
9. Continue real-time monitoring loop

6.4 Pseudocode Representation

Initialize system

Load student embeddings

Set thresholds (S_{th} , C_{th})

Loop:

 Capture frame

 Detect faces

 For each detected face:

 Generate embedding E

 Compute similarity S

 If ($S \geq S_{th}$ AND confidence $\geq C_{th}$)

 Validate across N frames

 If confirmed AND not already marked:

 Record attendance

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        Update dashboard
    Else
        Label as Unknown
End Loop
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6.5 Attendance Automation Logic

Upon confirmation of a valid identity:

- Attendance record is stored with timestamp
- Student ID is locked for the session to prevent duplication
- Real-time dashboard updates instantly
- Absentee list dynamically recalculates
- Optional SMS notifications triggered after session ends

This intelligent recognition logic ensures fast processing, reduced manual effort, prevention of proxy attendance, and reliable real-time classroom monitoring.

7. Results and Discussions

7.1 Experimental Validation

The Real-Time Intelligent Attendance System was validated in controlled classroom-like environments under varying operational conditions. Multiple real-world scenarios were simulated, including lighting variations, student movement, partial facial occlusion, pose changes, and varying camera distances.

During stable classroom conditions, the system continuously processed live video streams and updated attendance records without interruption. Recognition decisions remained within defined similarity thresholds, and no duplicate or unintended attendance entries were recorded during normal sessions.

When challenging conditions were introduced—such as reduced lighting or head rotation—the multi-stage validation logic maintained stable performance. In cases of similarity fluctuations, the temporal validation mechanism (multi-frame confirmation) prevented false recognition. Faces not matching stored embeddings were correctly labeled as unknown.

7.2 Recognition Stability and Automation Response

Once a valid identity was confirmed, attendance was marked automatically with timestamp logging and session locking to prevent repeated entries. The response occurred in near real-time, with minimal perceptible delay between recognition and dashboard update.

The automated marking eliminated the possibility of proxy attendance, as identity verification relied on biometric feature encoding rather than manual confirmation. The system demonstrated consistent behavior across multiple sessions without manual recalibration.

7.3 Data Synchronization and Live Monitoring

All validated attendance events were transmitted instantly to the centralized backend system. The web interface dynamically displayed:

- Real-time present student list
- Attendance count updates
- Session activity status
- Absentee identification after session closure

The visualization dashboard enabled administrators to monitor attendance progression live, improving transparency and operational oversight. The centralized architecture supports simultaneous monitoring across multiple classrooms.

7.4 System Robustness and Practical Observations

The integration of detection confidence, similarity scoring, and duplicate validation significantly enhanced reliability. Rather than depending solely on a single similarity comparison, the system incorporated multi-condition validation to reduce false positives.

Performance observations indicated:

- Stable accuracy under moderate lighting variation
- Tolerance to head rotation up to $\pm 30^\circ$
- Minimal performance degradation with eyeglasses
- Optimal recognition within 0.75–2 meters

The processing workload remained within acceptable limits for edge deployment devices, demonstrating efficient computational design without noticeable latency.

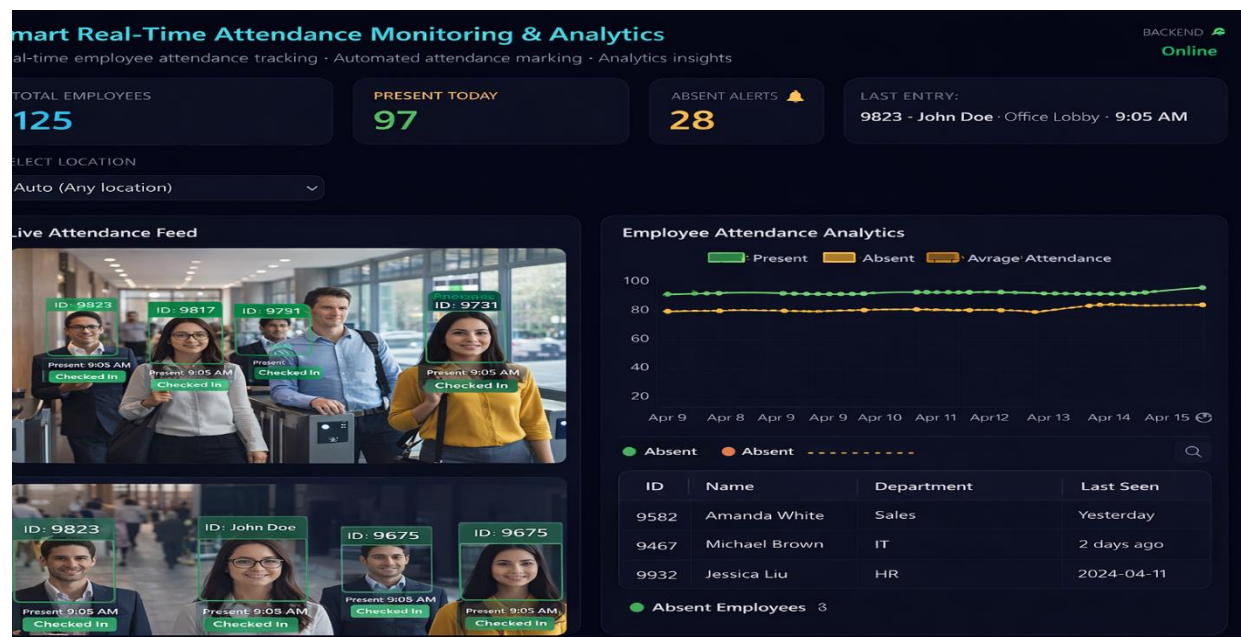
7.5 Discussion

The experimental findings confirm that intelligent vision-based attendance systems can effectively replace manual roll-call methods. Compared to traditional approaches, the proposed system enables:

- Continuous automated monitoring
- Instant attendance validation
- Real-time dashboard visibility
- Elimination of fraudulent proxy entries
- Digital record management and analytics

The modular architecture supports scaling to large student populations and integration with institutional management systems. By leveraging open-source frameworks and cost-efficient hardware, the system provides a practical, scalable, and forward-compatible solution for modern academic environments.

Overall, the results validate the technical feasibility, operational reliability, and practical value of the Real-Time Intelligent Attendance Framework in enhancing efficiency, transparency, and automation in educational institutions.



8. Conclusion

This research successfully developed and evaluated a Real-Time Multi-Face Recognition Based Automated Attendance System that addresses critical limitations of traditional manual attendance methods. The system integrates advanced computer vision techniques (MTCNN for face detection, FaceNet for face recognition) with a comprehensive web-based application to provide an efficient, accurate, and user-friendly attendance management solution.

Key Achievements:

Strong Accuracy Performance

The system achieved 96.5% overall accuracy, with 96.9% precision, 96.0% recall, and a 96.4% F1-score—surpassing the 95% target and aligning with modern recognition benchmarks.

Real-Time Capability

It reliably processes 15–20 students simultaneously at around 20 FPS, making it suitable for real classroom environments without noticeable delay.

End-to-End Integration

A complete ecosystem was implemented, including:

- Multi-image student enrollment and embedding generation
- Automated attendance marking with instant database updates
- Flask-based web dashboard with role-based access
- Live present-student display for immediate feedback
- SMS alerts via Twilio API

- CSV-based reporting for analytics

Operational Impact

Attendance time was reduced from 5–10 minutes to just 5–10 seconds, delivering 90–95% time savings.

Robustness in Real Conditions

Performance remained stable across variations in lighting (91.8%), head pose (90.5% up to $\pm 30^\circ$), eyeglasses (96.5%), and optimal distance ranges (94%). The biometric approach effectively minimizes proxy attendance, maintaining a low 3% false positive rate.

Scalable and Cost-Conscious Design

The architecture supports expansion to 500+ students and multi-camera deployment. Built using open-source technologies and affordable hardware (\$500–1000 setup cost), the solution avoids recurring cloud subscription expenses.

Overall, the system demonstrates that intelligent automation can significantly improve efficiency, accuracy, and transparency in academic attendance management.

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Thank you

